WHITEMATRIX SOFTWARE SOLUTIONS

MACHINE TEST ASSIGNMENT

AI Full Stack Engineer

CM KERALA – CHIEF MINISTER'S PUBLIC GRIEVANCE REDRESSAL PORTAL

SUBMISSION DEADLINE	MANDATORY SUBMISSION
12 November 2026 **8:00 PM IST**	Hosted application link and GitHub repository
DELIVERABLES	ASSESSMENT
Working prototype, documentation and demo video	Frontend, backend, database, APIs, AI and end-to-end workflow

THE SCENARIO

Imagine that you are the Chief Minister of Kerala. Citizens across the state face issues involving public infrastructure, healthcare, education, public services and other matters. You want to establish a dedicated online portal through which citizens can submit grievances directly to the Chief Minister's Office (CMO), track progress and receive updates.

Your task is to build a working prototype that enables the CMO to review grievances, forward them to the appropriate departments, monitor action and communicate outcomes to citizens. Every candidate must build this same application.

The final hosted application link and GitHub repository must be submitted no later than 12 November 2026 at 8:00 PM IST. Deploy your application and share the working hosted link before the deadline. Do not wait until the final minutes.

1. PUBLIC WEBSITE

Develop the following public-facing pages:

- **Home:** Portal introduction, navigation, prominent Submit Grievance and Track Grievance actions, sample statistics, featured departments and footer.
- **About Us:** Purpose, vision, mission and objectives of the grievance initiative.
- **How It Works:** Explain grievance submission, CMO review, department assignment, action and resolution.
- **Departments:** Display sample government departments with names and short descriptions.
- **Contact Us:** Contact form with name, email, subject and message; save enquiries in the database.
- **FAQ:** Frequently asked questions about submitting and tracking grievances.

2. CITIZEN PORTAL

- **Registration and login:** Register with name, email, mobile number and password; support login and logout.
- **Submit grievance:** Capture district, category, subject, detailed description and optional supporting image/document.
- **Reference number:** Generate a unique grievance ID after successful submission.
- **Track grievance:** Search by reference number and view current status, assigned department and status history.
- **Citizen dashboard:** Show submitted grievances, pending/resolved counts and previous complaint details.
- **Updates and feedback:** Display CMO/officer remarks and allow feedback after resolution.

3. CHIEF MINISTER'S OFFICE (CMO) DASHBOARD

The CMO dashboard is the central administrative module. An authorized administrator must be able to:

- View all grievances submitted by citizens.
- See total, pending, resolved and overdue grievance counts.
- Filter grievances by district, category, priority and status.
- Open individual grievance records and view attachments.
- Assign or forward grievances to the relevant government department/officer.
- Add internal administrative remarks and escalate delayed grievances.
- Monitor department responses and review action-taken reports.
- View district-wise and department-wise statistics using charts or tables.

4. DEPARTMENT OFFICER DASHBOARD

- Secure officer login using assigned credentials.
- View grievances forwarded by the CMO to that department.
- Review complaint details and supporting attachments.
- Update grievance status and add investigation remarks.
- Upload or enter an action-taken report.
- Submit resolution details back to the CMO for review.

5. MANDATORY GRIEVANCE WORKFLOW

Citizen submits grievance → System generates reference ID → CMO reviews → CMO forwards to department → Officer investigates and updates progress → Department submits action-taken report → CMO reviews response → Citizen receives update → Grievance is resolved or reopened.

Use these statuses: **Submitted, Under Review, In Progress, Resolved, Reopened.** Store every status change with its timestamp and the responsible user. The grievance must be traceable throughout its lifecycle.

6. MANDATORY AI INTEGRATION

Implement an **AI Grievance Assistant**. When a citizen enters a complaint description, the system should:

- Analyse the complaint text.
- Suggest the relevant department and grievance category.
- Recommend a priority level: Low, Medium or High.
- Generate a short summary for the CMO.
- Optionally flag potentially duplicate complaints for human review.

AI suggestions must be reviewable by the citizen or CMO administrator. AI must not independently make official decisions. Use an actual AI model/API where possible. If unavailable, clearly document the limitation and provide a labelled fallback; a static keyword classifier alone does not count as completed AI integration.

7. TECHNICAL REQUIREMENTS

Layer	Requirement
Frontend	React.js / Next.js or equivalent
Backend	Node.js + Express or equivalent
Database	MongoDB / MySQL / PostgreSQL or equivalent
API	REST APIs or equivalent service interface
Authentication	Role-based access for citizen, CMO admin and department officer

Layer	Requirement
AI	LLM/model API integration for grievance analysis
File handling	Functional upload and retrieval of supporting files
Dashboard	Charts/statistics based on stored application data

Minimum working API operations: citizen registration/login; submit grievance; retrieve/list grievances; track a grievance by ID; update grievance status; retrieve departments; retrieve CMO statistics; AI grievance analysis. Equivalent endpoint names and architecture are acceptable if documented.

8. DATABASE REQUIREMENTS

The database must support at minimum the following entities/tables or equivalent collections:

- Users (citizens, CMO administrators and department officers)
- Departments, Grievances and Grievance History
- Department Assignments, Attachments and Citizen Feedback
- Contact Enquiries

Persist submitted grievances and status history in the database. Use fictional sample data for demonstration.

9. MANDATORY DOCUMENTATION

Create a **DOCUMENTATION** folder in the GitHub repository containing:

File	Required contents
01_Project_Overview.md	Problem statement, objectives, scope and features
02_System_Architecture.md	Frontend, backend, database, APIs and AI architecture
03_Database_Design.md	Schema, entities/tables and relationships
04_API_Documentation.md	Endpoints, methods, parameters and request/response examples
05_AI_Integration.md	Model/API, prompt or workflow, limitations and safeguards
06_Setup_and_Testing.md	Installation, configuration, environment variables and test cases
07_User_Guide.md	How citizens, CMO administrators and officers use the system

Also include a root-level **README.md**, an **.env.example** file without real secrets, a system architecture diagram and a database ER diagram.

10. EVALUATION – 100 MARKS

Evaluation area	Marks
Frontend and UI/UX	10
Backend engineering and business logic	20
Database design	10
API development and integration	15
AI implementation	15
Grievance workflow functionality	15
Documentation	10
Security and testing	5

Evaluation area	Marks
TOTAL	100

11. DEVELOPMENT PRIORITIES

- Plan the solution and architecture; design the database.
- Build the public website and citizen-facing pages.
- Implement backend, database, authentication and APIs.
- Complete citizen grievance submission, tracking and dashboards.
- Implement CMO and department officer workflows.
- Integrate the AI grievance assistant and test the end-to-end process.
- Complete documentation, diagrams, deployment and hosted-link verification.

Prioritize a working end-to-end grievance workflow over optional features.

12. SUBMISSION REQUIREMENTS

- **Hosted application link – mandatory:** Deploy the application and send the working URL to the recruitment team before **12 November 2026, 8:00 PM IST**. The link must be accessible to evaluators without requiring them to run the code locally.
- GitHub repository link with complete source code and meaningful project structure.
- Completed DOCUMENTATION folder and required diagrams.
- 3–5 minute screen-recorded demonstration showing the main workflows.
- Environment configuration example with no passwords, tokens or API keys.
- Include demo login credentials needed by evaluators in the submission message or README. Use test accounts only.

13. DEADLINE. RULES AND IMPORTANT NOTES

FINAL SUBMISSION DEADLINE: 12 NOVEMBER 2026, 8:00 PM IST. Submissions received after the deadline may not be considered. Candidates are strongly advised to deploy and share their hosted link ahead of time, rather than waiting until the final minutes.

- AI coding assistants and open-source libraries are permitted. Disclose the tools and third-party services used.
- Paid APIs or subscriptions are not compulsory.
- Never commit credentials or secrets to GitHub.
- Copied projects will not qualify. Candidates must explain their implementation during the technical interview.
- Use fictional sample data only. Do not collect real citizen grievances or use official government credentials.
- Clearly identify the application as a recruitment prototype. Do not imply authorization or endorsement by the Government of Kerala or the Chief Minister's Office.

OUR EXPECTATION

At WhiteMatrix, we believe engineering is not just about writing code. It is about understanding a real problem, designing a practical solution and delivering working technology.

Imagine the responsibility of the Chief Minister. Build a digital system that helps the Chief Minister's Office listen to citizens, route grievances to the right departments and monitor action transparently.

ONE ASSIGNMENT. BUILD A WORKING SOLUTION.

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